



Exercises session 5

Class NP

1 Definitions

Definition 1.1 (Verifier) A **verifier** for a language L is a deterministic Turing machine V such that

$$L = \{w \mid V \text{ accepts } \langle w, c \rangle \text{ for some string } c\}$$

A **polynomial time verifier** V runs in time that is polynomial in w . A language is **polynomially verifiable** if it has a polynomial time verifier.

Definition 1.2 (NTIME)

$$NTIME(t(n)) = \{L \mid L \text{ is a language decided by a } O(t(n)) \text{ time non-deterministic TM}\}$$

Definition 1.3 (Class NP) Let's see definitions of the class NP.

1. NP is the class of all languages that have polynomial time verifiers.
2. Defining NP via NTMs

$$NP = \bigcup_k NTIME(n^k)$$

Theorem 1 A language is in NP (has a poly-time verifier) if and only if it is decided by some nondeterministic Turing machine running in polynomial time.

Definition 1.4 (Kleene star operation) Let A be a language. The operation **star** $(*)$ is defined as follows:

$$A^* = \{x_1x_2 \dots x_k \mid k \geq 0 \text{ and each } x_i \in A \text{ where } 1 \leq i \leq k\}$$

2 Session exercises

1. Prove that the class **P** is closed under intersection, union, concatenation and complement.
2. Prove that the class **NP** is closed under union, intersection, kleene star and concatenation.
3. In previous sessions, we have discussed about the following problems

- **Long path**

$\text{LPATH} = \{ \langle G, a, b, k \rangle : \text{there's a simple path from } a \text{ to } b \text{ of length at least } k \}$

- **Subset sum**

$\text{SUBSET-SUM} = \{ \langle A, k \rangle \mid \text{there exists } B \subseteq A \text{ such that } \sum_{b \in B} b = k \}$

Prove that these problems are in NP using both the non-deterministic turing machine and polynomial verifier definition.

4. Given a graph $G = (V, E)$, an **independent set** of G is a set $X \subseteq V$ such that for all $(u, v) \in X \times X$, $(u, v) \notin E$.
 - Give an example of a graph and two non-empty sets of vertices, such that one is an independent set and the other is not
 - Prove that the following problem is in NP

$L = \{ \langle G, k \rangle \mid G \text{ has an independent set of size } k \}$